NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project Title: AI-Based Smart Quiz System.

Course Code: CSE 4204
Course Title: Mobile Computing Lab

Submitted By:

Team Name : CSE4204-8A-T03
Section : 8A

GitHub Repository: https://github.com/sudipto224/AI-based-Smart-Quiz-System

Submitted to:

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Designation: Assistant Professor
Department of CSE

Date of Submission: 19/08/2026

TEAM INFORMATION

Official Team Name	CSE4204-8A-T03
Section	8A
Project Title	AI-Based Smart Quiz System
GitHub Repository	https://github.com/sudipto224/AI-based-Smart-Quiz-System

Team Information:

Role	Full Name	Student ID
Team Leader + Backend	Sudipto Das	11220320830
Artificial Intelligence	Rakibur Rahman Mishan	11220320829
Database + Frontend	Aditi Roy	11220320846

Summary of Completed Features

No.	Completed Feature	Implementation Summary
1	User authentication	Registration, login, logout, password hashing, session management and protected routes are integrated through Laravel Breeze.
2	Role-based access	Teacher and student behavior is separated using the is_teacher role flag and protected route/controller checks.
3	Teacher course management	Teachers can create, view, edit and delete their own courses and configure time per question.
4	Question management	Teachers can add, edit and delete MCQs with four options, one correct answer and an explanation.
5	Course-aware AI generation	The teacher selects a course, enters a topic, and the backend sends a strict course-aware prompt to Groq. The generated MCQ auto-fills the existing question form for human review.
6	Material-based AI generation	Teachers can upload PDF/PPTX learning material, request 5 or 10 questions, preview the generated items, remove unwanted questions and save approved items to the existing question bank.
7	Timed quiz workflow	Students can start a course quiz, answer questions under the configured timer and rely on automatic submission when time expires.
8	Result and explanation	Quiz attempts store selected answers and show score, selected answer, correct answer and explanation in the detailed result review.
9	Leaderboard	Course-wise rankings use quiz attempt data and performance/time information.
10	Suspicious attempt detection	Tab switching and unusually fast answering are detected using rule-based heuristics; suspicious

No.	Completed Feature	Implementation Summary
		attempts are stored for teacher review.
11	Database integration	Users, courses, questions, sessions and quiz attempts are persisted in MySQL through Laravel/Eloquent.
12	Quizora branding and UI consistency	The default Laravel branding has been replaced by a reusable Quizora logo/name across the landing, authentication and authenticated layouts, including favicon and browser title.
13	AI verification logging	Laravel logs the requested and served Groq model so the team can verify the exact model used during generation.

Partially Completed Features and Remaining Features

Item	Status	Reason / Current State	Planned Completion
Cross-device responsive QA	In Progress	Main layouts are responsive, but final checks on multiple phone/tablet widths are still being completed.	18 Aug 2026
Formal error-scenario test matrix	In Progress	Application-level validation and AI error handling exist; structured testing of database/network/API failure scenarios is still being documented.	18 Aug 2026
GitHub README and repository cleanup	In Progress	README and repository documentation need to be aligned with Quizora branding, current React/Inertia frontend, Groq platform and GPT-OSS 120B model.	18 Aug 2026
Formal	Not Started	Week 09 prepares the integrated build;	Next course

Item	Status	Reason / Current State	Planned Completion
testing/debugging phase		systematic testing is planned as the next project phase.	phase
Public live deployment	Not Started	A public live demo is optional and has not yet been deployed; the project currently runs locally.	After local testing

Frontend-Backend Integration Status

The major frontend components are connected to Laravel routes and controllers using Inertia.js and standard form submissions, while AI generation uses an Axios request from the React interface to the Laravel backend. The backend performs validation, authorization, database operations and AI communication before returning data to the frontend.

Integration Area	Status	Evidence / Behavior
Authentication requests	Integrated	Registration/login/logout requests are processed by Laravel and session state is shared with React/Inertia pages.
Course forms	Integrated	Create/edit/delete operations are validated in Laravel and persisted in MySQL.
Question forms	Integrated	Manual and AI-generated values use the same question storage workflow.
AI requests/responses	Integrated	React sends course_id + topic; Laravel builds the prompt, calls Groq, validates JSON and returns the MCQ.
Quiz submission	Integrated	Answers, score, timing and suspicious-attempt metadata are processed by the backend and

Integration Area	Status	Evidence / Behavior
		persisted.
Results/leaderboard	Integrated	Stored quiz_attempt data is retrieved for detailed result views and course ranking.
Error handling	Integrated / Improving	Validation, unauthorized access, AI errors and malformed AI responses are handled; broader failure testing is being finalized.

AI Integration Status

The AI feature is part of the real teacher workflow rather than a standalone demonstration. The teacher opens a course, enters a topic in the Add Question page, and requests a question. The browser sends only the course identifier and topic to Laravel; the Groq credential remains on the server. Laravel then loads the selected course, constructs a strict course-aware prompt, calls Groq, validates the response and returns a structured MCQ to the same form for teacher review.

AI Item	Current Status
AI Platform	Groq Cloud API
Current Model	openai/gpt-oss-120b
Primary AI Feature	Course-aware MCQ generation from selected course + teacher-entered topic
Additional AI Feature	Bulk quiz generation from PDF/PPTX course material with preview and approval
Response Format	Structured JSON: question, four options, correct_answer and explanation
Validation	Required fields, exactly four unique non-empty options, correct-answer matching, JSON validation and duplicate-question check
Failure Handling	User-friendly error response; manual question form remains

AI Item	Current Status
	available; no unrelated dummy question is inserted
Credential Security	API credential is loaded through environment configuration and is not intended to be committed to GitHub
Verification	Laravel log records both requested_model and served_model

Important classification: suspicious-attempt detection is rule-based/heuristic (tab switching and average answer time), not a machine-learning or LLM feature.

Authentication Status

Authentication Item	Status	Verification
Registration	Completed	New users can register and the account is stored in the users table.
Login	Completed	Existing users authenticate through Laravel Breeze.
Logout	Completed	Authenticated session is terminated through the logout route.
Password handling	Completed for local application	Passwords are hashed by Laravel; password-reset pages exist. Local mail is configured for development logging rather than public email delivery.
Session management	Completed	Database-backed sessions are used in the current environment.
Protected routes	Completed	Authenticated pages require login; teacher functionality also checks role/ownership.
Role-based access	Completed	Teacher and student paths are separated using is_teacher and controller authorization checks.

Major End-to-End User Workflows

Teacher Workflow :

- Teacher registers/logs in.
- Teacher opens the dashboard and selects or creates a course.
- Teacher manages the course question bank.
- Teacher may enter a topic and generate a course-aware AI MCQ, or upload course material for bulk generation.
- Laravel processes the request, validates the AI response and returns the generated content.
- Teacher reviews/edits the generated question and saves it.
- The approved question is stored in the MySQL questions table.
- Teacher can later review suspicious attempts for owned courses.

Student Workflow :

- Student registers/logs in.
- Student opens the student dashboard and selects an available course.
- Student starts a timed quiz.
- Answers are submitted manually or automatically when the timer expires.
- Laravel calculates the score, stores answer selections and attempt metadata, and evaluates rule-based suspicious behavior.
- The result page displays score, selected answers, correct answers and explanations.
- The attempt contributes to the course-wise leaderboard when eligible.

Major Problems Encountered and Solutions Implemented

Problem	**Cause**	**Solution**
AI generated unrelated Mobile Computing context for other courses.	The original prompt contained hard-coded Mobile Computing context and artificial seed/timestamp text.	Rebuilt AI generation around the currently selected course + topic; removed unrelated hard-coded context and random seed/timestamp instructions.
Quiz score was correct but Detailed Review displayed	Selected answers were used during scoring but	Added a JSON answers column, updated QuizAttempt fillable/casts,

Problem	Cause	Solution
'(Not answered)'.	were not persisted in quiz_attempts.	saved answers during submission and read them on the Result page.
Previous Groq model became unavailable/decommissioned.	The earlier Llama 3.3 70B model selection was retired by the provider.	Migrated the active Groq requests to openai/gpt-oss-120b and added model logging for verification.
Laravel default logo and old product naming remained on several pages.	Branding was duplicated across layouts/pages instead of coming from a shared component.	Introduced reusable Quizora Brand/ApplicationLogo components, updated layouts, favicon, app title and environment app name.
Frontend changes sometimes did not appear immediately.	Browser/build cache continued serving previously compiled Vite assets.	Rebuilt assets with npm run build, cleared Laravel optimization cache and performed a hard refresh.
AI response can be malformed, empty or duplicated.	External LLM output is not guaranteed to match the application contract.	Added strict JSON response requirements, server-side validation, exact option/correct-answer checks, duplicate checks and clear error handling.

Changes from Original Plan

Area	Original / Earlier Approach	Current Approach	Reason
Project branding	Generic 'AI-Based Smart Quiz System'	Quizora - AI-Based Smart Quiz	A fixed product identity, logo, favicon and reusable brand component improve consistency and professional presentation.

Area	Original / Earlier Approach	Current Approach	Reason
Frontend approach	Earlier plan/repository documentation referenced a simpler Laravel/custom frontend	React + Inertia.js + Tailwind CSS integrated with Laravel	Provides a cleaner component-based UI and smoother frontend-backend interaction.
AI provider/model	Original proposal/README referenced Google Gemini; Week 08 later used Groq with Llama 3.3 70B	Groq Cloud with openai/gpt-oss-120b	The implementation evolved during AI integration and the previous model was replaced after provider decommissioning.
AI course scope	AI generation initially focused on Mobile Computing	Course-aware generation for the currently selected course	Prevents cross-course content errors and makes the feature useful across the application.
AI failure behavior	Earlier fallback could return a dummy/local question	User-friendly error with manual question form preserved	Avoids inserting misleading or unrelated academic content.
Quiz content generation	Single topic-based question generation	Added PDF/PPTX material-based 5/10-question generation with preview	Provides meaningful additional value for real teacher content preparation.
Result persistence	Score stored, selected-answer review incomplete	Selected answers are now stored with quiz attempts	Allows accurate detailed result review.
Cheating detection description	Sometimes described as AI-enhanced	Explicitly documented as rule-based/heuristic	Keeps the technical description accurate and avoids overstating the AI component.

UI/UX and Error Handling Improvements

- Consistent Quizora logo and project name across landing, authentication and authenticated layouts.
- Browser favicon and page titles updated to the Quizora identity.
- Course-aware labels and helper text make AI generation clearer to teachers.
- Loading state is shown while an AI question is being generated.
- Success and error messages are returned in understandable language rather than raw API output.
- Unauthorized teacher/course access is blocked through ownership checks.
- AI errors, empty responses, invalid JSON, missing fields, duplicate options and duplicate questions are handled on the backend.
- Forms use validation and required fields instead of relying on static/hardcoded content.
- Responsive layout classes are used throughout; final device-width verification is part of remaining Week 09 QA.

Feature Completion Checklist

Major Feature	**Status**	**Reason if Incomplete**	**Planned Completion**
User registration	Completed	-	-
User login/logout	Completed	-	-
Role-based teacher/student access	Completed	-	-
Teacher course management	Completed	-	-
Teacher question management	Completed	-	-
Course-aware AI MCQ generation	Completed	-	-
PDF/PPTX material-based AI generation	Completed	-	-
Student course selection	Completed	-	-

Major Feature	Status	Reason if Incomplete	Planned Completion
Timed quiz and auto-submit	Completed	-	-
Stored answer review on result page	Completed	-	-
Course-wise leaderboard	Completed	-	-
Suspicious attempt detection/report	Completed	-	-
Quizora branding and shared logo	Completed	-	-
Model verification logging	Completed	-	-
Cross-device responsive verification	In Progress	Final checks on multiple phone/tablet widths.	18 Aug 2026
Structured failure-scenario test matrix	In Progress	Formal Week 09 verification is still being documented.	18 Aug 2026
README/current repository documentation	In Progress	Repository text needs current branding/model/stack information.	18 Aug 2026
Formal testing/debugging phase	Not Started	Planned after Week 09 integration milestone.	Next course phase
Public live deployment	Not Started	Optional; local build is currently used for review.	After local testing

GitHub Repository and Live Demo Status

Repository : https://github.com/sudipto224/AI-based-Smart-Quiz-System

The repository is used for semester-long version control and team collaboration. For Week 09, the latest implementation should be pushed with meaningful commit messages, the README should

be aligned with the current Quizora implementation, and unnecessary temporary files should be excluded. API keys, passwords and database credentials must remain outside version control.

Live demo status: no public deployment link is currently available. The integrated application is being demonstrated from the local Laravel development environment. A public deployment can be considered after the formal testing/debugging phase.

Current Project Screenshots

The following screenshots provide implementation evidence for the Week 09 review. One dashboard evidence slot is intentionally reserved because a current dashboard capture was not available among the supplied screenshots; replace that single image with the latest logged-in dashboard screenshot before exporting the final PDF.

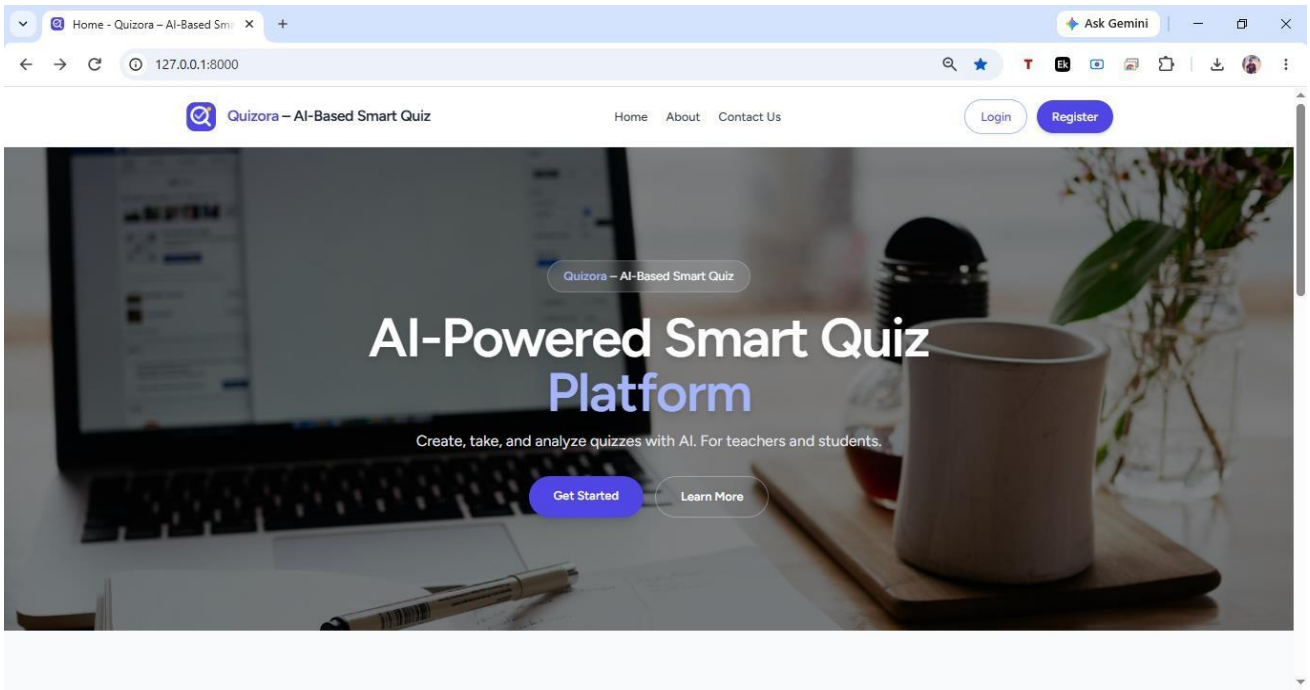


Figure 1(a). Quizora home page showing branding, navigation, and main call-to-action.

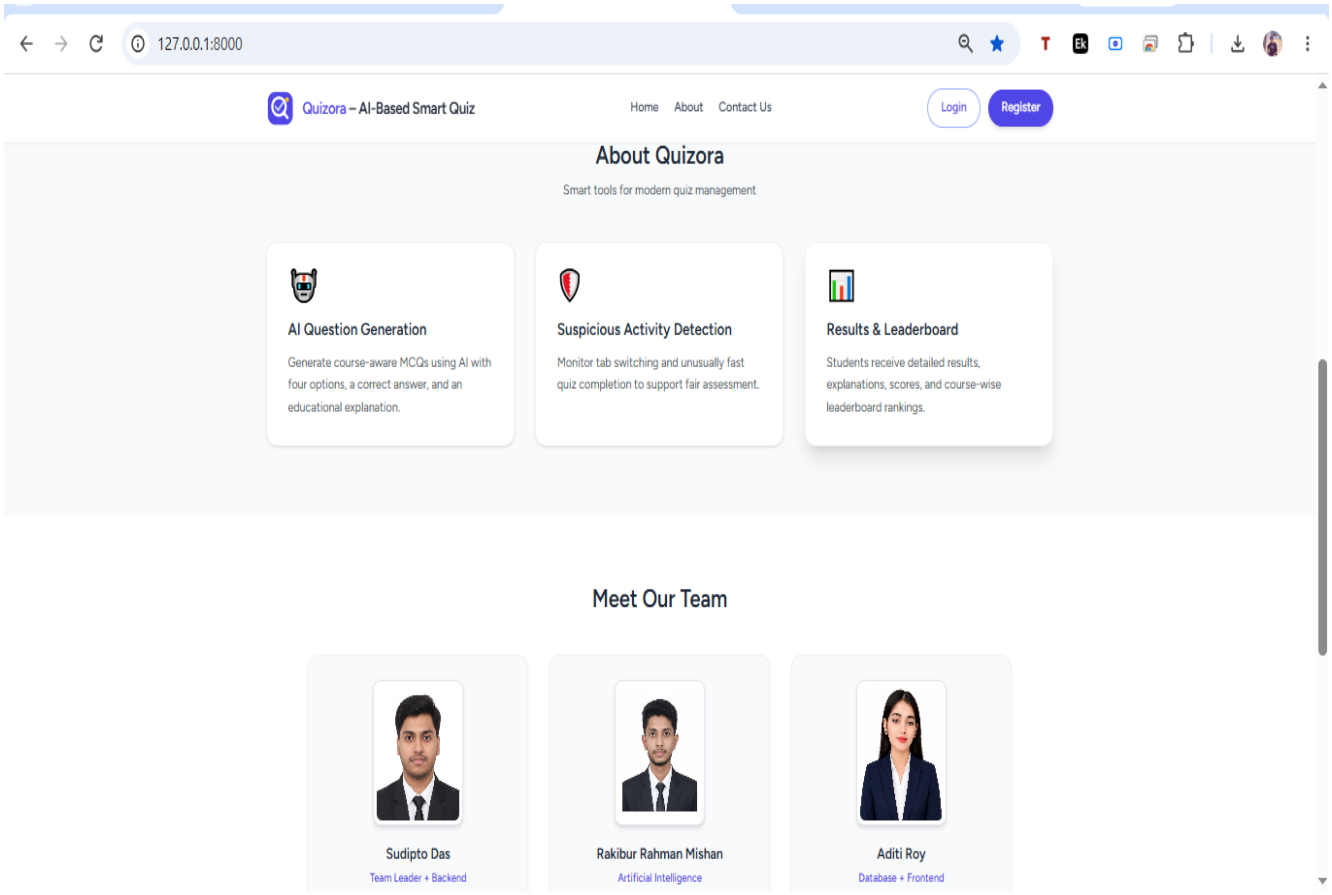


Figure 1(b). About and Team sections showing core features and team members.

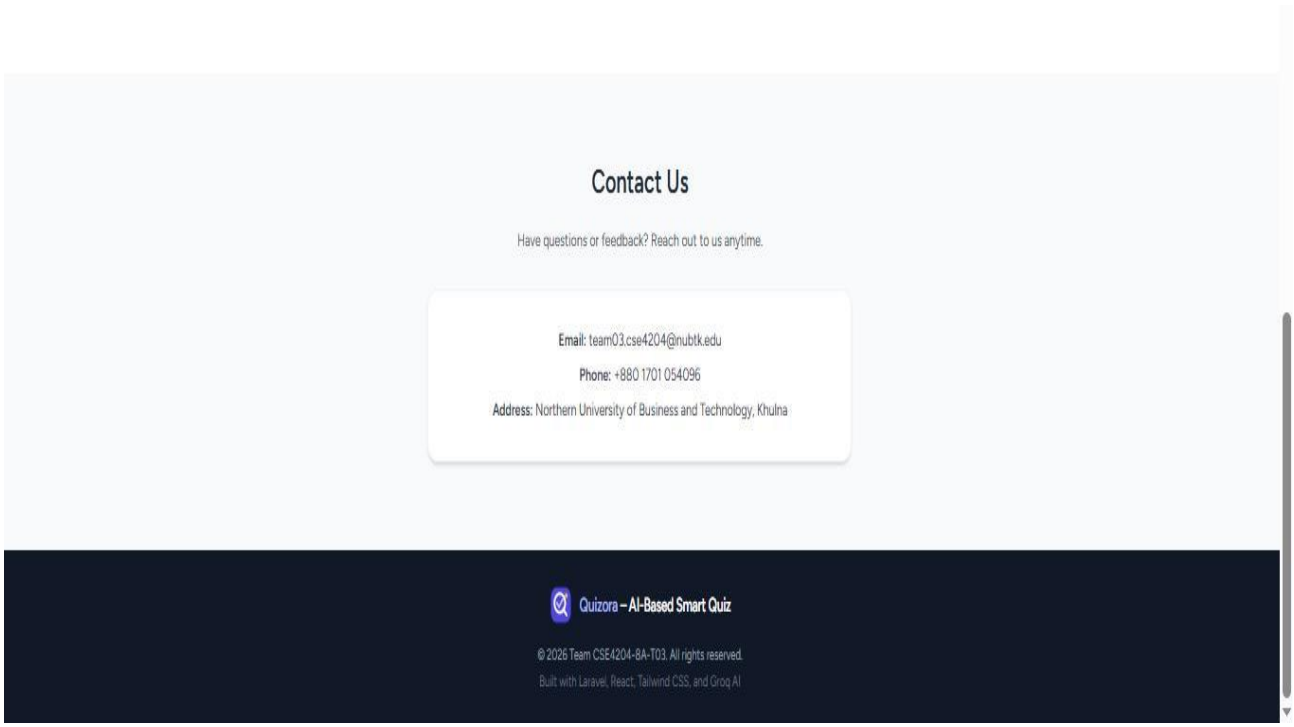


Figure 1(c). Contact and Footer sections showing contact information and Quizora branding.

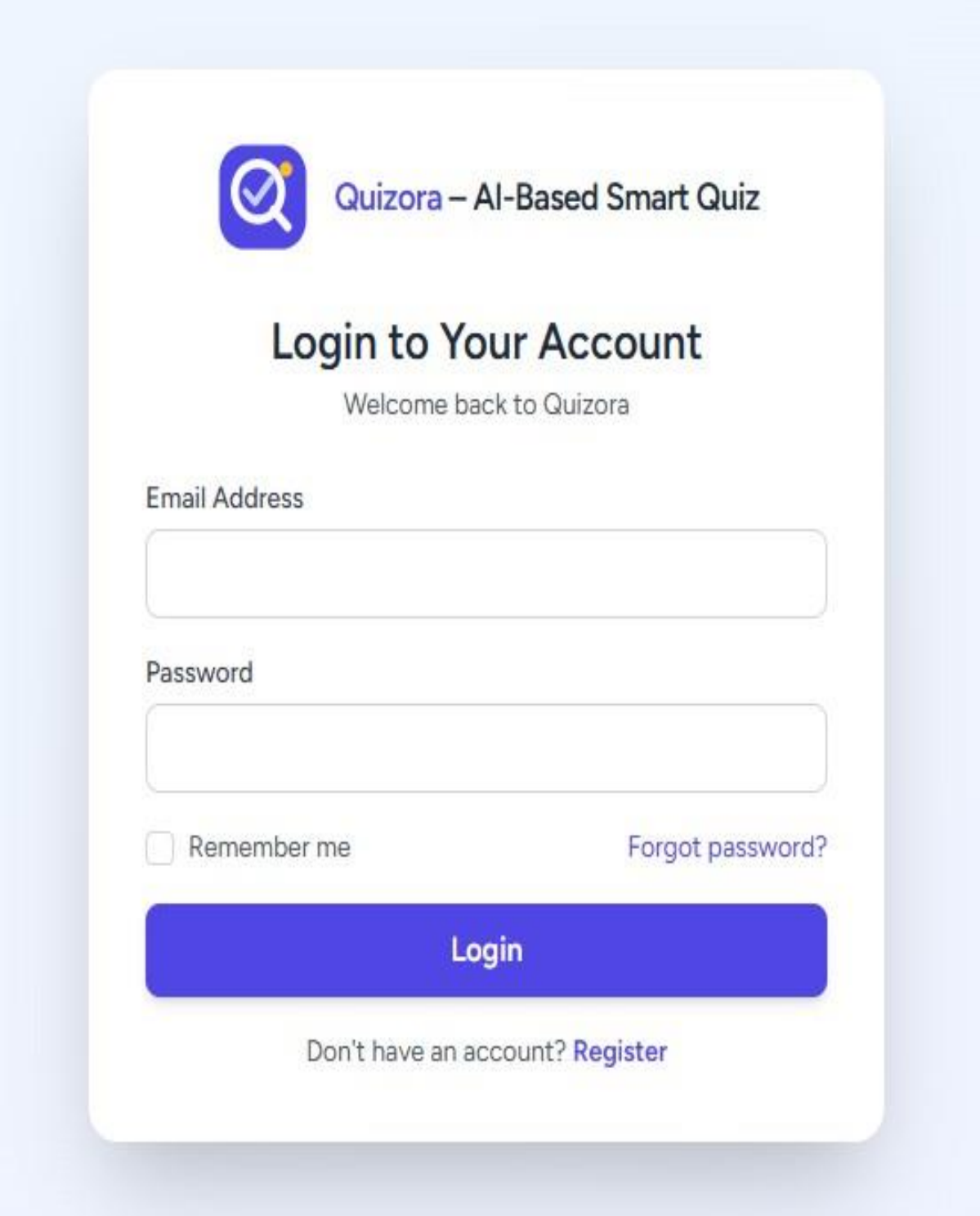


Figure 2(a). Login form and authentication interface.

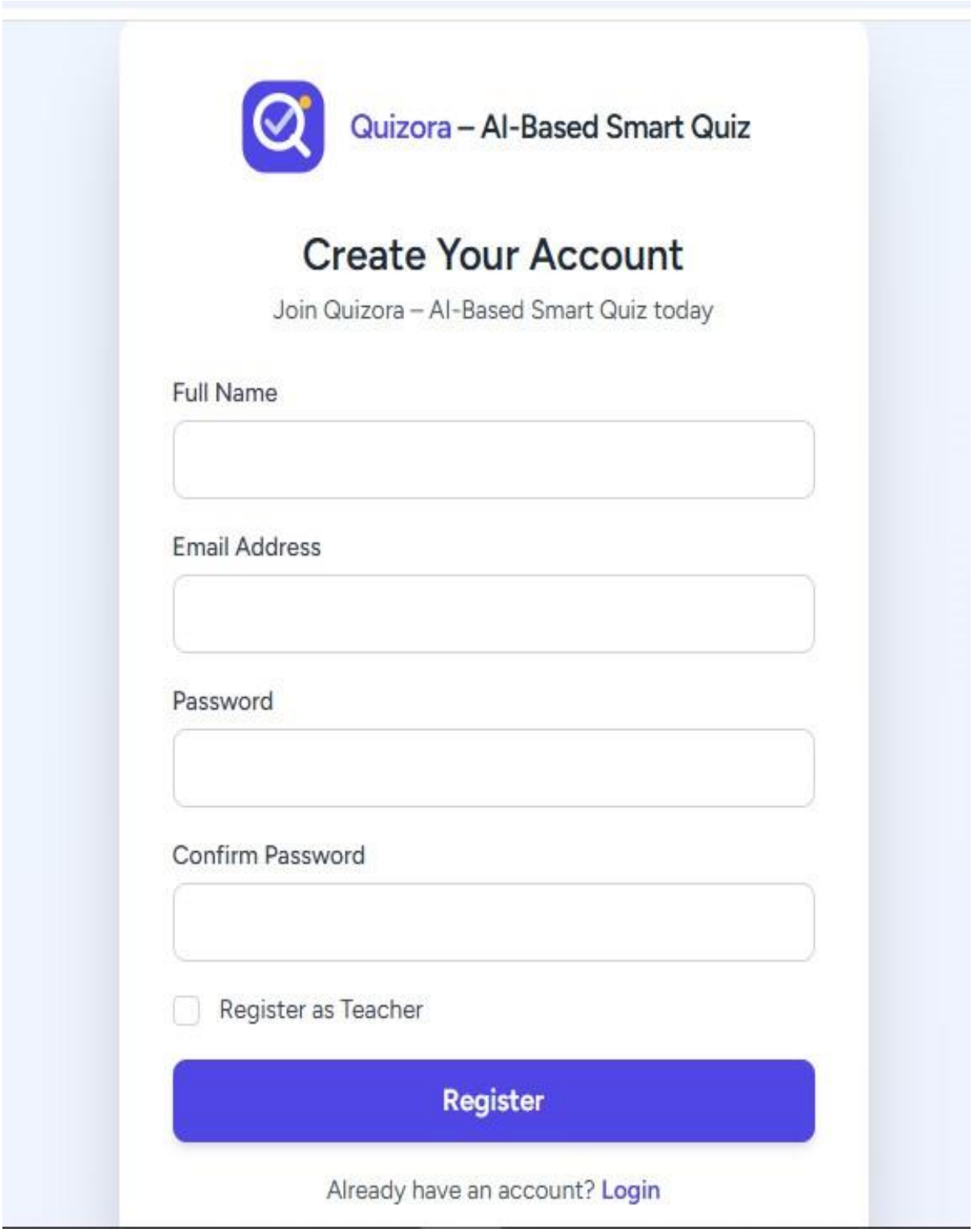


Figure 2(b). Registration form including the teacher-role selection option.

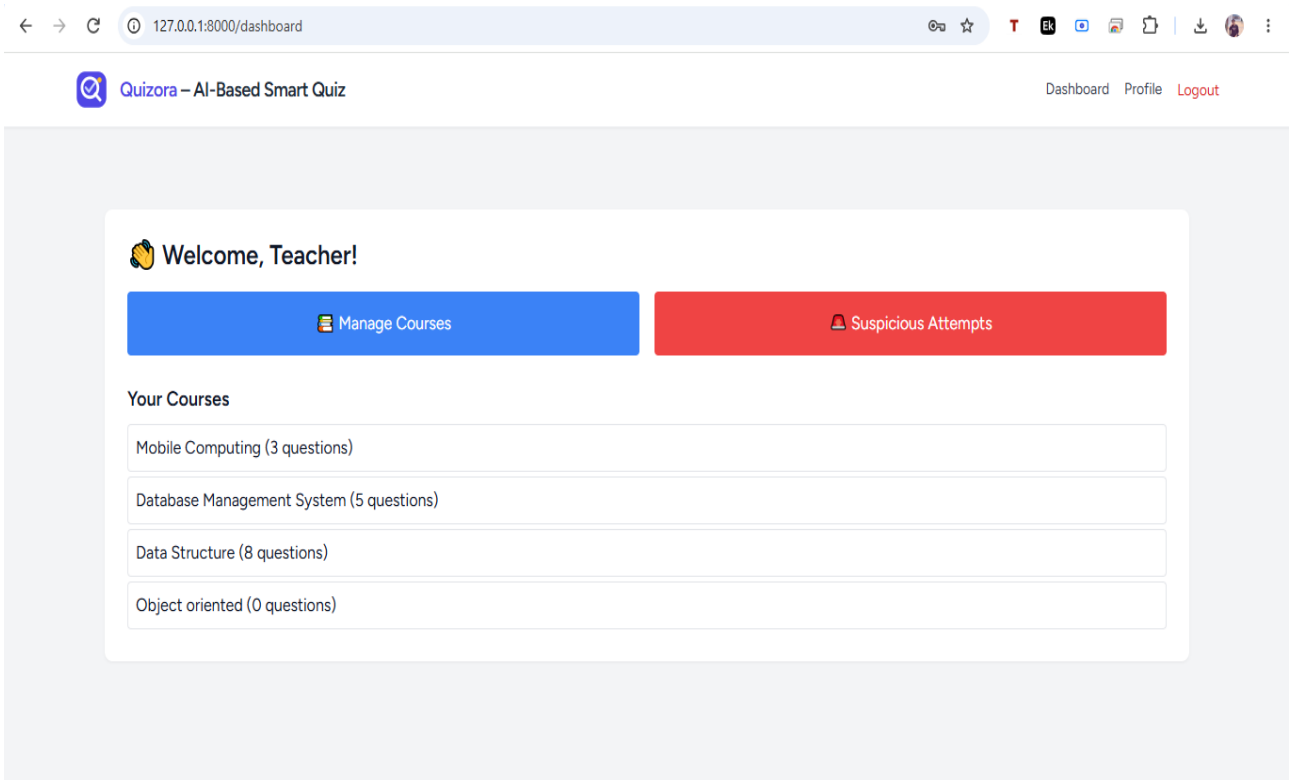


Figure 3(a). Teacher Dashboard showing course management, question counts, and suspicious attempt monitoring.

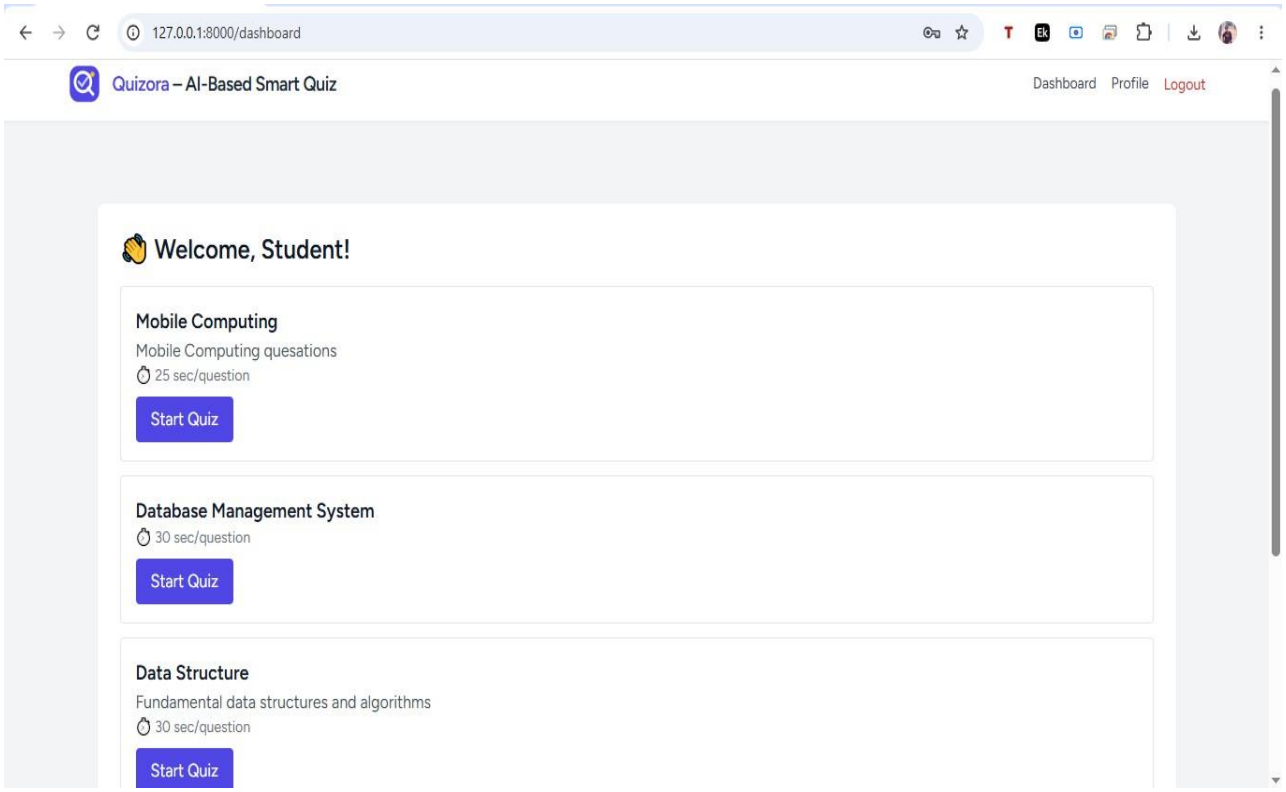


Figure 3(b). Student Dashboard showing available courses, quiz timing, and Start Quiz options.

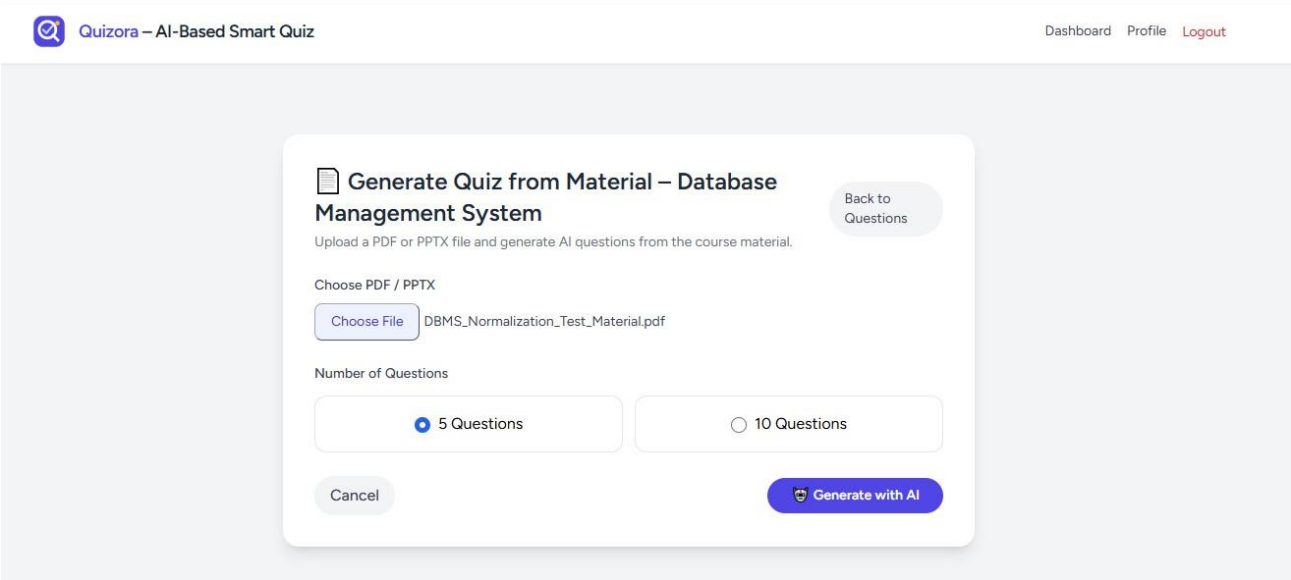


Figure 4(a). Material-based AI quiz generation interface for uploading PDF/PPTX files and selecting the number of questions.

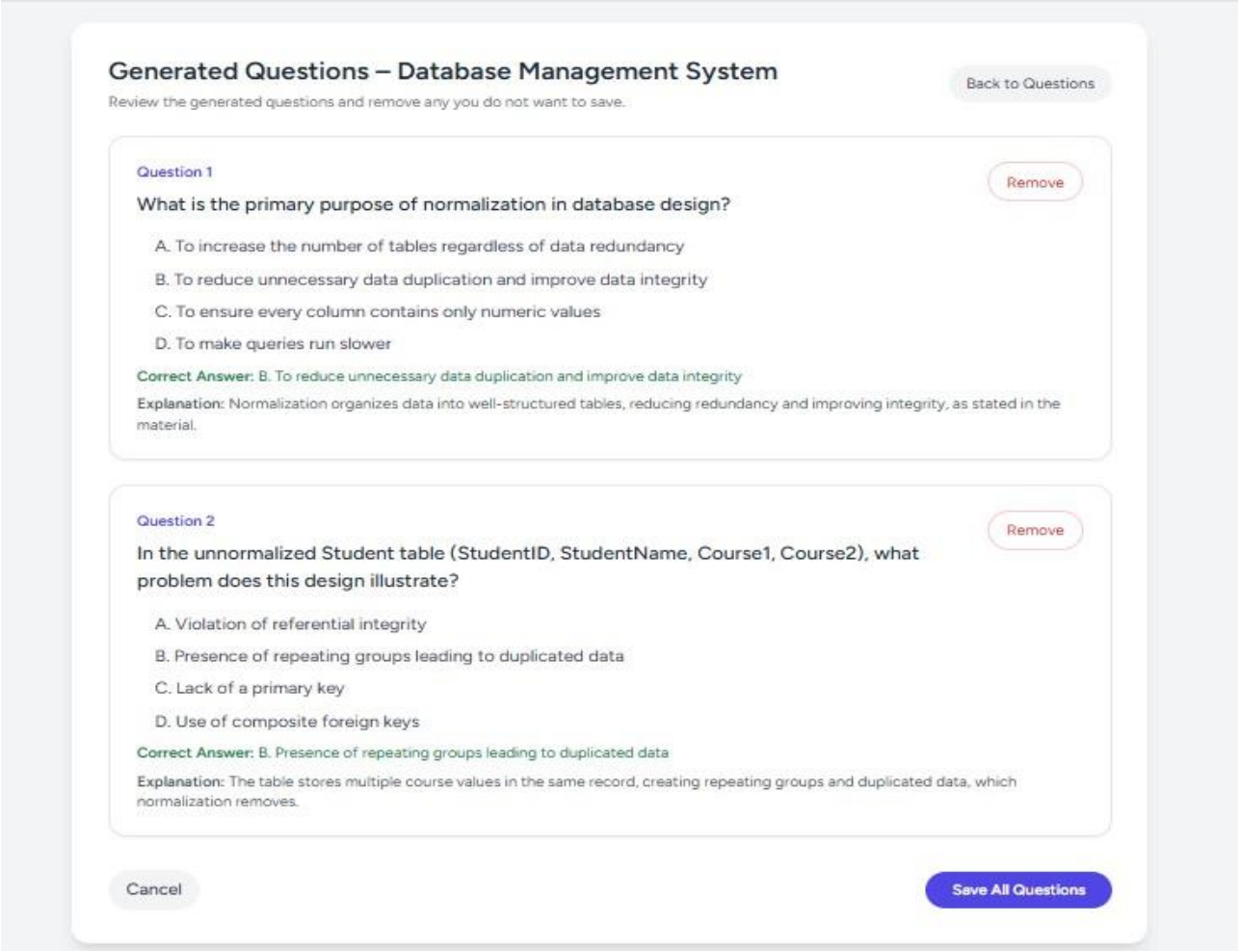


Figure 4(b). AI-generated question preview showing MCQs, correct answers, explanations, and Save All Questions option..

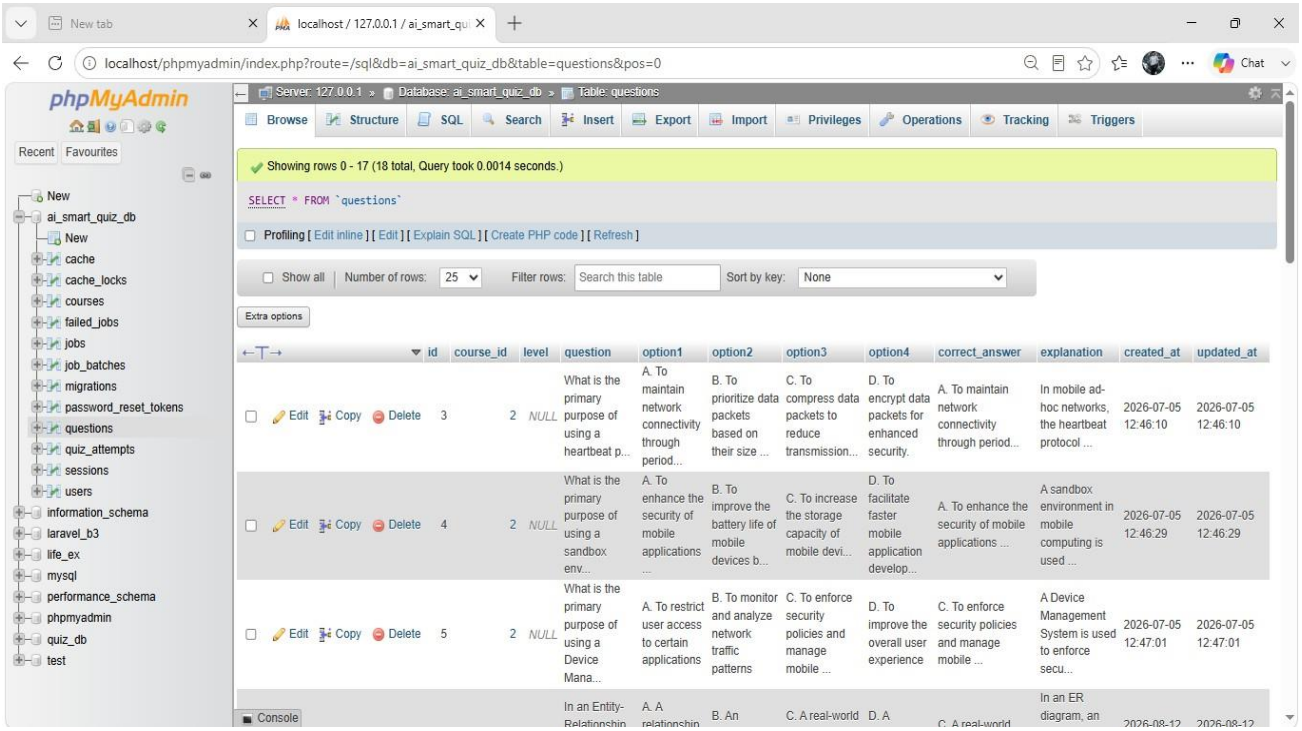


Figure 5. MySQL/phpMyAdmin evidence showing AI-generated question data stored in the questions table.

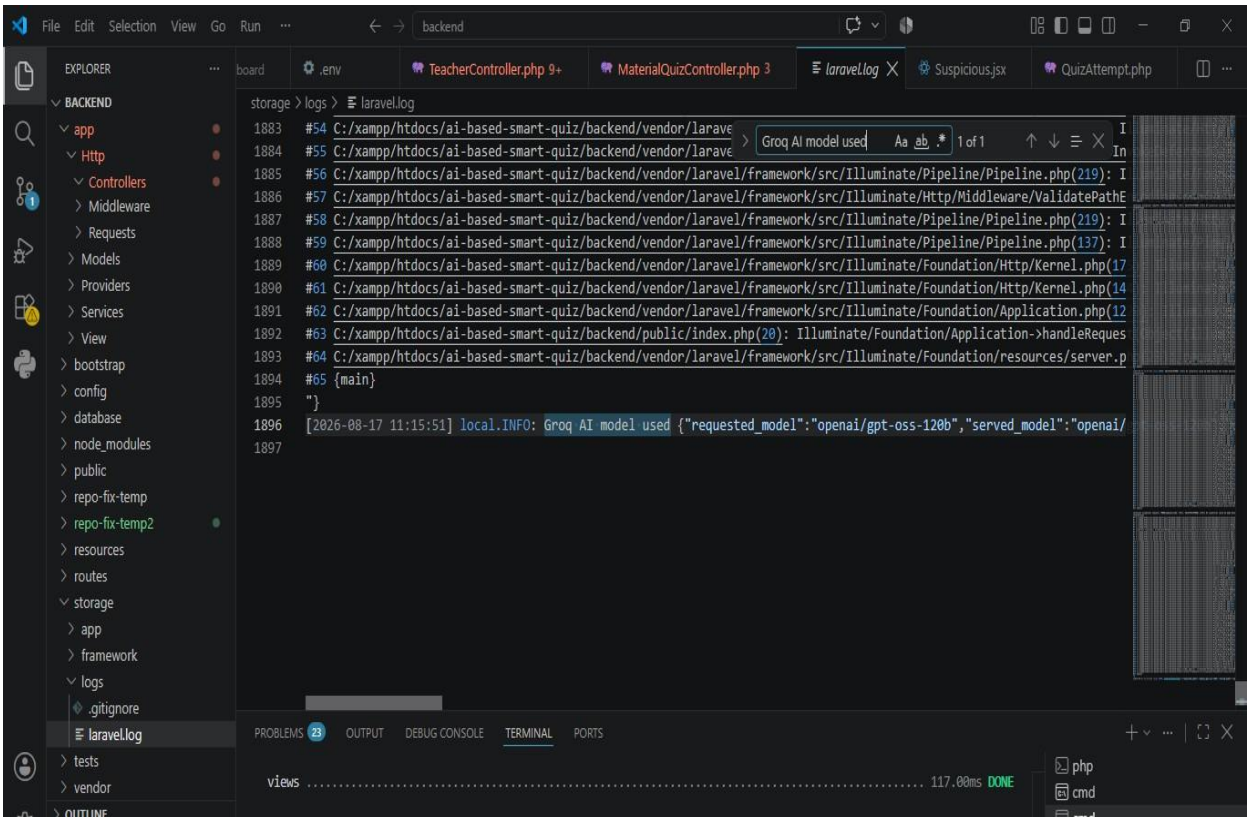


Figure 6. Laravel log evidence used to verify the requested and served Groq model.

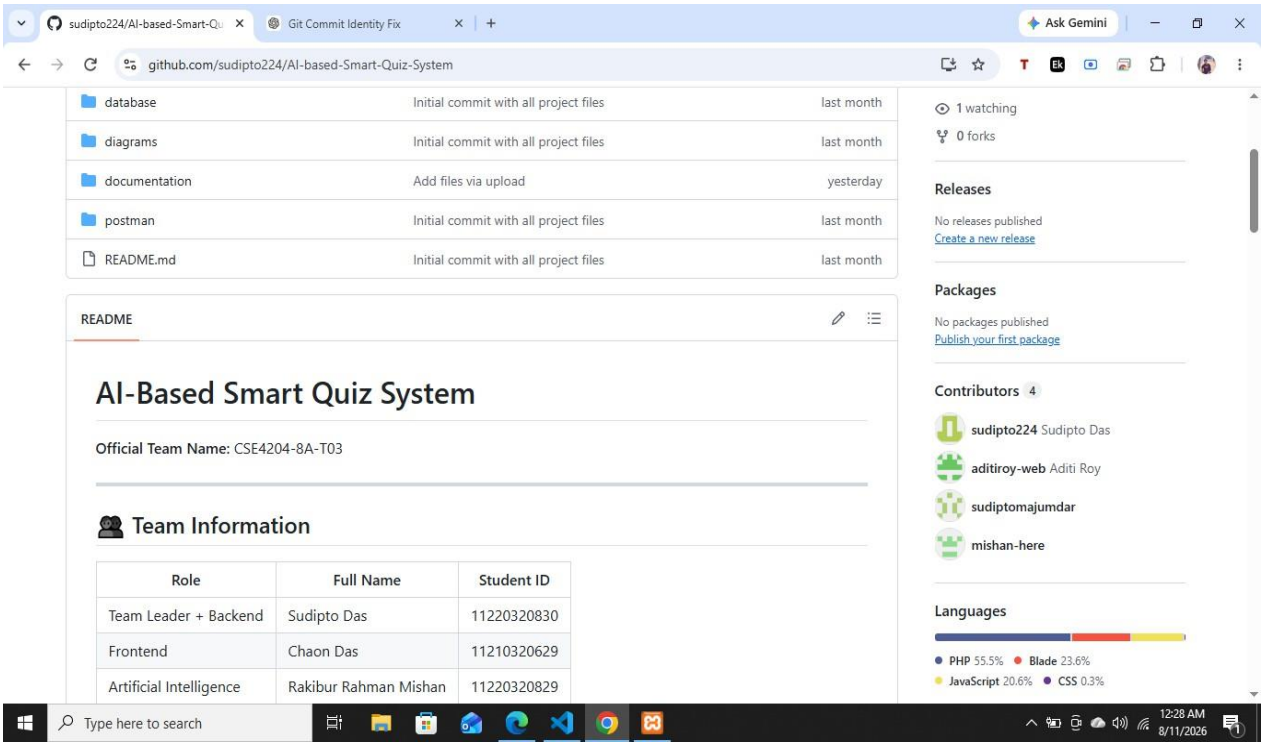


Figure 7. GitHub repository and README/repository structure evidence.

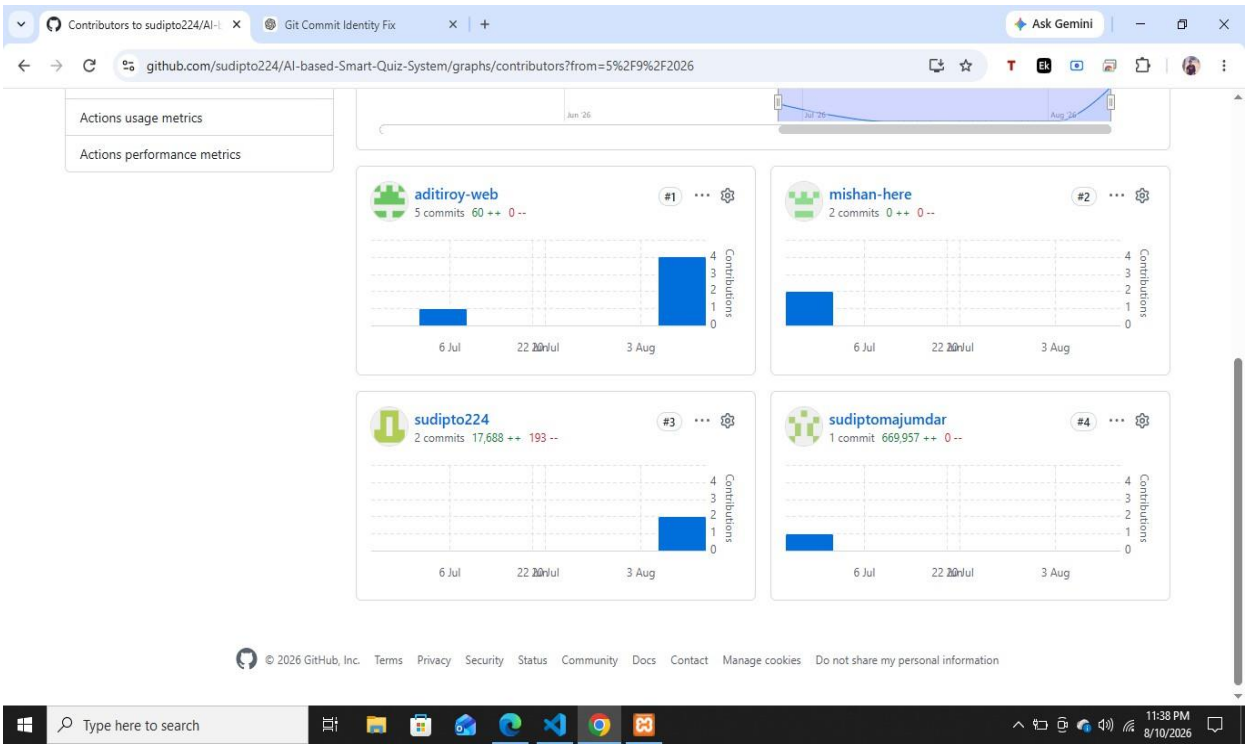


Figure 8. GitHub contributor activity evidence supporting continuous team development.

Conclusion

The Week 09 milestone shows that Quizora has moved beyond separate frontend, backend, database and AI modules into an integrated application. The major teacher and student workflows

are functional, real system data is used throughout the main workflow, AI generation is connected to course management, authentication and ownership controls are active, and result/database issues identified during integration have been fixed. The remaining tasks are mainly formal responsive/error testing, repository documentation cleanup and preparation for the next testing/debugging phase.

The system is therefore ready for the mid-project review, with the team able to demonstrate authentication, core teacher/student functions, database persistence, AI generation, integration behavior, and the current Quizora UI.